



Spectra of Poisson functionals and applications in continuum percolation

Joint with G. Peccati (Luxembourg) and D. Yogeshwaran (ISI Bangalore)

Chinmoy Bhattacharjee

June 21, 2024

Workshop on Stein's Method and Networks, University of Oxford

Voronoi percolation

Given a homogeneous *Poisson process* with **intensity 1** on $\mathbb{R}^2$, independently, colour each cell black or white with probabilities p and $1 - p$ (marked process η).

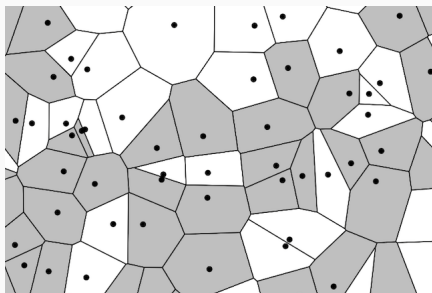


Figure 1: Voronoi percolation (from Bollobás and Riordan '06)

Left-right crossing of squares

Goal is to measure 'sensitivity to noise' : Does small perturbations in η affect percolative properties?

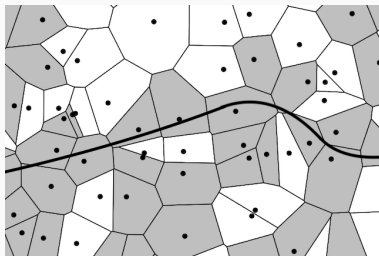
Left-right crossing of squares

Goal is to measure 'sensitivity to noise' : Does small perturbations in η affect percolative properties? Non-trivial at criticality : Fix $p = 1/2$.

Left-right crossing of squares

Goal is to measure 'sensitivity to noise' : Does small perturbations in η affect percolative properties? Non-trivial at criticality : Fix $p = 1/2$.

Let $f_L = \pm 1$ indicator of a L-R black crossing of $W_L = [-L, L]^2$:



Noise sensitivity

By symmetry

$$\begin{aligned} P_{1/2}(f_L(\eta) = 1) &= P_{1/2}(\text{L-R black crossing of } W_L) \\ &= P_{1/2}(\text{Top-Bottom white crossing of } W_L) = \frac{1}{2}. \end{aligned}$$

Noise sensitivity

By symmetry

$$\begin{aligned} \mathbf{P}_{1/2}(f_L(\eta) = 1) &= \mathbf{P}_{1/2}(\text{L-R black crossing of } W_L) \\ &= \mathbf{P}_{1/2}(\text{Top-Bottom white crossing of } W_L) = \frac{1}{2}. \end{aligned}$$

For η the original configuration and an ϵ -noisy version η^ϵ , the system is sensitive when

$$\mathbf{P}_{1/2}(f_L(\eta^\epsilon) = 1 | f_L(\eta) = 1) \approx \frac{1}{2} \approx \mathbf{P}_{1/2}(f_L(\eta^\epsilon) = 1 | f_L(\eta) = -1).$$

Noise sensitivity

By symmetry

$$\begin{aligned} \mathbf{P}_{1/2}(f_L(\eta) = 1) &= \mathbf{P}_{1/2}(\text{L-R black crossing of } W_L) \\ &= \mathbf{P}_{1/2}(\text{Top-Bottom white crossing of } W_L) = \frac{1}{2}. \end{aligned}$$

For η the original configuration and an ϵ -noisy version η^ϵ , the system is sensitive when

$$\mathbf{P}_{1/2}(f_L(\eta^\epsilon) = 1 | f_L(\eta) = 1) \approx \frac{1}{2} \approx \mathbf{P}_{1/2}(f_L(\eta^\epsilon) = 1 | f_L(\eta) = -1).$$

Definition: $(f_L)_{L>0}$ is **noise sensitive** (NS) under the dynamics η^ϵ if $\forall \epsilon > 0$,

$$\lim_{L \rightarrow \infty} \text{Cov}(f_L(\eta), f_L(\eta^\epsilon)) = 0.$$

Frozen dynamics

- Sample the unperturbed random configuration according to $\eta \subset \mathbb{R}^2 \times \{\pm 1\}$.
- Resample the colour of each cell independently, at rate 1 to obtain the ε -noisy version η^ε after time $\varepsilon > 0$.

Frozen dynamics

- Sample the unperturbed random configuration according to $\eta \subset \mathbb{R}^2 \times \{\pm 1\}$.
- Resample the colour of each cell independently, at rate 1 to obtain the ε -noisy version η^ε after time $\varepsilon > 0$.

Sharp NS under frozen dynamics (Vanneuille '21)

For the critical Voronoi percolation under the frozen dynamics, there exists $A_L \rightarrow \infty$ such that

$$\lim_{L \rightarrow \infty} \text{Cov}(f_L(\eta), f_L(\eta^{\varepsilon_L})) = 0 \quad \text{when } \varepsilon_L A_L \rightarrow \infty,$$

and the limit is at least $c > 0$ for any sequence ε_L with $\varepsilon_L A_L \rightarrow 0$.

Frozen dynamics

- Sample the unperturbed random configuration according to $\eta \subset \mathbb{R}^2 \times \{\pm 1\}$.
- Resample the colour of each cell independently, at rate 1 to obtain the ε -noisy version η^ε after time $\varepsilon > 0$.

Sharp NS under frozen dynamics (Vanneuville '21)

For the critical Voronoi percolation under the frozen dynamics, there exists $A_L \rightarrow \infty$ such that

$$\lim_{L \rightarrow \infty} \text{Cov}(f_L(\eta), f_L(\eta^{\varepsilon_L})) = 0 \quad \text{when } \varepsilon_L A_L \rightarrow \infty,$$

and the limit is at least $c > 0$ for any sequence ε_L with $\varepsilon_L A_L \rightarrow 0$.

Frozen dynamics is conditionally a dynamics on $\{\pm 1\}^{\mathbb{Z}^2}$. Garban, Pete and Schramm '10 obtained Sharp NS via spectral samples.

Given marked configuration η , noisy version η^ε is given by

$$\eta^\varepsilon = \eta_1 + \eta_2, \quad t > 0,$$

- η_1 a **thinning** of η [delete each point independently w.p. $(1 - e^{-\varepsilon})$],
- η_2 an **independent Poisson process** with intensity $(1 - e^{-\varepsilon})$.

Ornstein Uhlenbeck dynamics

Given marked configuration η , noisy version η^ε is given by

$$\eta^\varepsilon = \eta_1 + \eta_2, \quad t > 0,$$

- η_1 a **thinning** of η [delete each point independently w.p. $(1 - e^{-\varepsilon})$],
- η_2 an **independent Poisson process** with intensity $(1 - e^{-\varepsilon})$.

Not possible to reduce to a discrete dynamics; only non-sharp results by Ahlberg, Baldasso '18; Last, Peccati, Yogeshwaran '23.

Ornstein Uhlenbeck dynamics

Given marked configuration η , noisy version η^ε is given by

$$\eta^\varepsilon = \eta_1 + \eta_2, \quad t > 0,$$

- η_1 a **thinning** of η [delete each point independently w.p. $(1 - e^{-\varepsilon})$],
- η_2 an **independent Poisson process** with intensity $(1 - e^{-\varepsilon})$.

Not possible to reduce to a discrete dynamics; only non-sharp results by Ahlberg, Baldasso '18; Last, Peccati, Yogeshwaran '23.

Motivation: Show sharp results for the OU dynamics.

Sharp noise sensitivity under OU dynamics

Sharp NS under OU dynamics (B., Peccati, Yogeswaran '24+)

Under the OU dynamics, for $A_L \rightarrow \infty$ as before, $(f_L)_{L>0}$ exhibits **sharp noise sensitivity** at scale $1/A_L$, i.e.,

$$\lim_{L \rightarrow \infty} \text{Cov}(f_L(\eta), f_L(\eta^{\varepsilon_L})) = 0 \quad \text{when } \varepsilon_L A_L \rightarrow \infty,$$

and the limit is at least $c > 0$ for any sequence ε_L with $\varepsilon_L A_L \rightarrow 0$.

Sharp noise sensitivity under OU dynamics

Sharp NS under OU dynamics (B., Peccati, Yogeswaran '24+)

Under the OU dynamics, for $A_L \rightarrow \infty$ as before, $(f_L)_{L>0}$ exhibits **sharp noise sensitivity** at scale $1/A_L$, i.e.,

$$\lim_{L \rightarrow \infty} \text{Cov}(f_L(\eta), f_L(\eta^{\varepsilon_L})) = 0 \quad \text{when } \varepsilon_L A_L \rightarrow \infty,$$

and the limit is at least $c > 0$ for any sequence ε_L with $\varepsilon_L A_L \rightarrow 0$.

We benefit from the sharp NS result by Vanneuville '21, via a covariance comparison between the Frozen and the OU dynamics.

Sharp noise sensitivity under OU dynamics

Sharp NS under OU dynamics (B., Peccati, Yogeswaran '24+)

Under the OU dynamics, for $A_L \rightarrow \infty$ as before, $(f_L)_{L>0}$ exhibits **sharp noise sensitivity** at scale $1/A_L$, i.e.,

$$\lim_{L \rightarrow \infty} \text{Cov}(f_L(\eta), f_L(\eta^{\varepsilon_L})) = 0 \quad \text{when } \varepsilon_L A_L \rightarrow \infty,$$

and the limit is at least $c > 0$ for any sequence ε_L with $\varepsilon_L A_L \rightarrow 0$.

We benefit from the sharp NS result by Vanneuville '21, via a covariance comparison between the Frozen and the OU dynamics.

For the proof of no noise sensitivity, we introduce the new notion of **spectral point process**; a continuum counterpart of spectral samples.

Let η be a homogeneous Poisson process with intensity $\lambda > 0$ in $\mathbb{R}^2$.
The black/occupied region is given by $\cup_{x \in \eta} B_1(x)$.

Let η be a homogeneous Poisson process with intensity $\lambda > 0$ in $\mathbb{R}^2$.
The black/occupied region is given by $\cup_{x \in \eta} B_1(x)$.

$\exists$ critical parameter $\lambda_c > 0$, such that for $\lambda > \lambda_c$, $\exists$ a unique unbounded occupied component.

Boolean percolation

Let η be a homogeneous Poisson process with intensity $\lambda > 0$ in $\mathbb{R}^2$. The black/occupied region is given by $\cup_{x \in \eta} B_1(x)$.

$\exists$ critical parameter $\lambda_c > 0$, such that for $\lambda > \lambda_c$, $\exists$ a unique unbounded occupied component.

Fix $\lambda = \lambda_c$. Consider the L-R occupied crossing functionals $f_L \in \{\pm 1\}$ of $W_L = [-L, L]^2$ as before.

Boolean percolation

Let η be a homogeneous Poisson process with intensity $\lambda > 0$ in $\mathbb{R}^2$. The black/occupied region is given by $\cup_{x \in \eta} B_1(x)$.

$\exists$ critical parameter $\lambda_c > 0$, such that for $\lambda > \lambda_c$, $\exists$ a unique unbounded occupied component.

Fix $\lambda = \lambda_c$. Consider the L-R occupied crossing functionals $f_L \in \{\pm 1\}$ of $W_L = [-L, L]^2$ as before.

Non-sharp results for noise sensitivity in Boolean percolation in Ahlberg et. al. 14.

Noise stability

A second approach to study the sensitivity of a system to random noise can be via measuring its [stability](#).

A second approach to study the sensitivity of a system to random noise can be via measuring its [stability](#).

Definition: $(f_L)_{L>0}$ is [noise stable](#) if

$$\lim_{\varepsilon \rightarrow 0} \sup_L \mathbf{P} \{f_L(\eta) \neq f_L(\eta^\varepsilon)\} = 0.$$

A second approach to study the sensitivity of a system to random noise can be via measuring its **stability**.

Definition: $(f_L)_{L>0}$ is **noise stable** if

$$\lim_{\epsilon \rightarrow 0} \sup_L \mathbf{P} \{f_L(\eta) \neq f_L(\eta^\epsilon)\} = 0.$$

Compared to noise sensitivity, captures the opposite situation : $f_L(\eta^\epsilon)$ almost completely determines $f_L(\eta)$ if ϵ is **small**, uniformly in L .

Sharp noise stability in Boolean percolation under OU dynamics (B., Peccati, Yogeshwaran '24+)

There exists $A_L \rightarrow \infty$ such that under the OU dynamics, $(f_L)_{L>0}$ is **noise stable** for any sequence ε_L with $\varepsilon_L A_L \rightarrow 0$, i.e.,

$$\lim_{L \rightarrow \infty} \mathbf{P} \{f_L(\eta) \neq f_L(\eta^{\varepsilon_L})\} = 0.$$

On the other hand, if $\varepsilon_L A_L \rightarrow \infty$, then it is **not noise stable**, i.e.,

$$\liminf_{L \rightarrow \infty} \mathbf{P} \{f_L(\eta) \neq f_L(\eta^{\varepsilon_L})\} > 0.$$

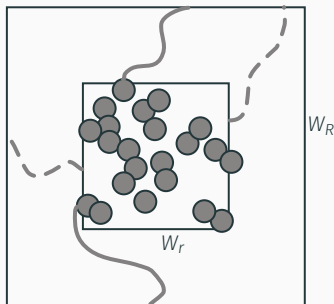
4-arm probability : Quasi-multiplicativity

In all results, $A_L = L^2 \alpha_4(1, L) \rightarrow \infty$, where α_4 is the 4-arm probability.

4-arm probability : Quasi-multiplicativity

In all results, $A_L = L^2 \alpha_4(1, L) \rightarrow \infty$, where α_4 is the 4-arm probability.

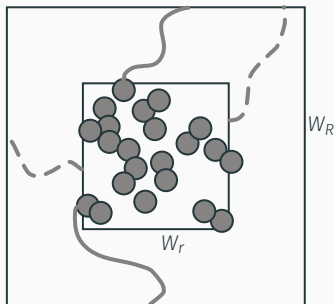
For $1 \leq r \leq R < \infty$, $\alpha_4(r, R)$ is the probability of the event



4-arm probability : Quasi-multiplicativity

In all results, $A_L = L^2 \alpha_4(1, L) \rightarrow \infty$, where α_4 is the 4-arm probability.

For $1 \leq r \leq R < \infty$, $\alpha_4(r, R)$ is the probability of the event



Quasi-multiplicativity of α_4 (B., Peccati, Yogeshwaran '24+)

In critical Boolean percolation,

$$\alpha_4(r_1, r_3) \asymp \alpha_4(r_1, r_2) \alpha_4(r_2, r_3), \quad 1 \leq r_1 \leq r_2 \leq r_3.$$

Main tool : Concentration of spectral mass

The *level of noise sensitivity* of a Boolean function is naturally encoded in its **Fourier spectrum**. *High frequency* implies sensitivity, while *low frequency* implies stability.

Main tool : Concentration of spectral mass

The *level of noise sensitivity* of a Boolean function is naturally encoded in its **Fourier spectrum**. *High frequency* implies sensitivity, while *low frequency* implies stability.

The crossing functional $f_L(\eta)$ admits a unique *chaotic decomposition*

$$f_L(\eta) = \sum_{k=0}^{\infty} I_k(u_k),$$

where I_k and u_k are the **multiple Wiener-Itô integral** of order k , and the **k -th kernel** in the decomposition, respectively.

Main tool : Concentration of spectral mass

The *level of noise sensitivity* of a Boolean function is naturally encoded in its **Fourier spectrum**. *High frequency* implies sensitivity, while *low frequency* implies stability.

The crossing functional $f_L(\eta)$ admits a unique *chaotic decomposition*

$$f_L(\eta) = \sum_{k=0}^{\infty} I_k(u_k),$$

where I_k and u_k are the **multiple Wiener-Itô integral** of order k , and the **k -th kernel** in the decomposition, respectively. In particular,

$$1 = \mathbb{E}[f_L(\eta)^2] = \sum_{k=0}^{\infty} k! \|u_k\|^2.$$

Main tool : Concentration of spectral mass

The *level of noise sensitivity* of a Boolean function is naturally encoded in its **Fourier spectrum**. *High frequency* implies sensitivity, while *low frequency* implies stability.

The crossing functional $f_L(\eta)$ admits a unique *chaotic decomposition*

$$f_L(\eta) = \sum_{k=0}^{\infty} I_k(u_k),$$

where I_k and u_k are the **multiple Wiener-Itô integral** of order k , and the **k -th kernel** in the decomposition, respectively. In particular,

$$1 = \mathbb{E}[f_L(\eta)^2] = \sum_{k=0}^{\infty} k! \|u_k\|^2.$$

For N_L with $\mathbf{P}\{N_L = k\} = k! \|u_k\|^2$, sharp noise sensitivity/stability follows by showing concentration of N_L around $\mathbf{E}N_L \asymp L^2 \alpha_4(1, L)$.

Spectral point process

To estimate the first two moments of N_L , we construct in a natural way a [spectral point process](#) with N_L points.

Spectral point process

To estimate the first two moments of N_L , we construct in a natural way a **spectral point process** with N_L points.

- Given $N_L = k$, obtain the random vector $(X_1^{(k)}, \dots, X_k^{(k)}) \in \mathbb{X}^k$ following a probability density proportional to

$$(x_1, \dots, x_k) \mapsto u_k^2(x_1, \dots, x_k).$$

Spectral point process

To estimate the first two moments of N_L , we construct in a natural way a **spectral point process** with N_L points.

- Given $N_L = k$, obtain the random vector $(X_1^{(k)}, \dots, X_k^{(k)}) \in \mathbb{X}^k$ following a probability density proportional to

$$(x_1, \dots, x_k) \mapsto u_k^2(x_1, \dots, x_k).$$

- Define the **spectral process** as

$$\gamma_L(B) := \sum_{i=1}^{N_L} \delta_{X_i^{(N_L)}}(B), \quad B \in \mathcal{X}.$$

Spectral point process

To estimate the first two moments of N_L , we construct in a natural way a **spectral point process** with N_L points.

- Given $N_L = k$, obtain the random vector $(X_1^{(k)}, \dots, X_k^{(k)}) \in \mathbb{X}^k$ following a probability density proportional to

$$(x_1, \dots, x_k) \mapsto u_k^2(x_1, \dots, x_k).$$

- Define the **spectral process** as

$$\gamma_L(B) := \sum_{i=1}^{N_L} \delta_{X_i^{(N_L)}}(B), \quad B \in \mathcal{X}.$$

The size of the spectral process can be neatly related to the number of **Pivotal points**, i.e., those $x \in \eta$ such that $D_x^- f_L(\eta) \neq 0$.

Further directions

It is typically easier to prove sharp noise stability. Showing sharp noise sensitivity under OU dynamics requires a precise understanding the lower tail of N_L :

It is typically easier to prove sharp noise stability. Showing sharp noise sensitivity under OU dynamics requires a precise understanding the lower tail of N_L : to conclude noise sensitivity for $\varepsilon_L \mathbf{E} N_L \rightarrow \infty$, it suffices to show that

$$\lim_{c \rightarrow 0} \liminf_{L \rightarrow \infty} \mathbf{P} \{N_L \geq c \mathbf{E} N_L\} = 1.$$

It is typically easier to prove sharp noise stability. Showing sharp noise sensitivity under OU dynamics requires a precise understanding the lower tail of N_L : to conclude noise sensitivity for $\varepsilon_L \mathbf{E} N_L \rightarrow \infty$, it suffices to show that

$$\lim_{c \rightarrow 0} \liminf_{L \rightarrow \infty} \mathbf{P} \{N_L \geq c \mathbf{E} N_L\} = 1.$$

Explore other avenues such as the [differential approach](#) by Tassion, Vanneuville '20 in continuum percolation.

Thank you!